

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Lecture 4 Animations: Student Guide

KKKC6353 Nanoelectronic Devices — learn by experimenting

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How to use this guide

Six interactive animations go with the Lecture 4 slides and notes. They are at **faizverse.github.io/nanoelectronic-devices/L4** and work on a laptop or phone, online or saved for offline use.

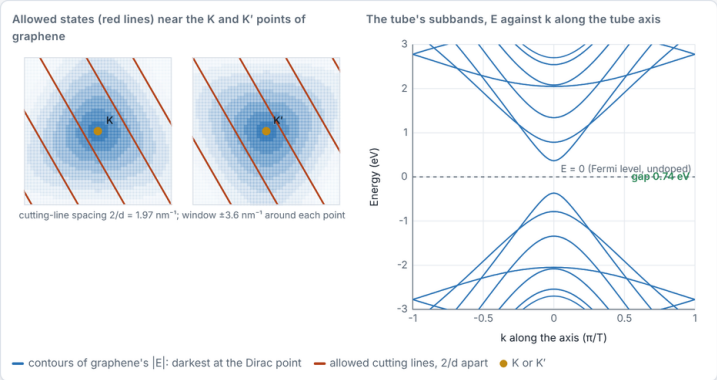
For each animation:

1. **Read first.** Skim the notes section listed, then read *What it shows* below. The page opens in a simple view; press **More detail** when you want the numbers and graphs.
2. **Predict, try, explain.** Before you move a slider, write down what you expect. Then try it. If the result surprised you, work out why.
3. **Answer the questions** at the end of each page in this guide, then check your answers under *Check yourself* on the web page.
4. **Note the big idea** in one sentence of your own.

Allow about 10 minutes per animation. Animation 1 is optional: it shows where the nanotube facts in the Part 1 recap come from. Animation 6 has two tabs and no More detail view.

#	Animation	Read first	Link
1	Nanotube band structure: zone folding	Doc 1, § 4–5	cnt-band-structure
2	The ballistic nanotube transistor	Doc 2, § 3–4	cntfet-ballistic
3	Schottky-barrier nanotube transistor	Doc 2, § 2 and 4.4	sb-cntfet
4	FinFET and nanosheet: width and control	Doc 3, § 1–2	finfet-nanosheet
5	SOI: partially and fully depleted	Doc 3, § 6–7	soi-pd-fd
6	Compound semiconductors: speed and colour	Doc 4, § 1–2	compound-semiconductors

1 Nanotube band structure: zone folding



Link: https://faizverse.github.io/nanoelectronic-devices/L4/cnt-band-structure.html
Read first: Document 1, Sections 4–5

What it shows. Choose how the graphene sheet is rolled, (n, m), and watch the allowed states cut through the Dirac cone of graphene. See why one tube in three is metallic, how the band gap scales as 1/d, and where the sharp peaks in the density of states come from.

Left. Near each corner of graphene's Brillouin zone (K and K′) the π bands form a cone, drawn here as contours: the darker the blue, the closer to zero energy. Rolling the sheet into a tube makes the wavefunction repeat around the circumference, so only wavevectors on parallel **cutting lines** survive, 2/d apart.

Right. Each cutting line is one subband of the tube. If a line passes exactly through K, two subbands cross at E = 0 and the tube is **metallic**. Otherwise the nearest line misses K by one third of the spacing, and a gap opens: $E_g = 2a_{cc}\gamma_0/d$.

More detail adds the density of states, whose sharp van Hove peaks sit at each subband edge, and the gap of every tube between 0.6 and 2.5 nm.

> **The big idea.** Rolling graphene into a tube lets only certain electron waves fit around it. If one of them passes through the point where graphene's bands touch, the tube is a metal; otherwise it has a gap, and thinner tubes have bigger gaps.

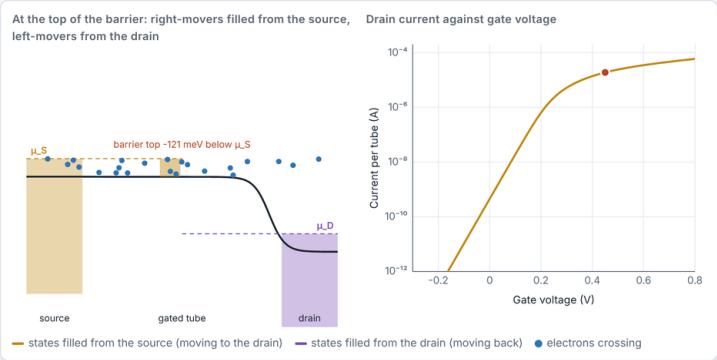
Try this

1. **Predict** which of (9,0), (10,0) and (11,0) is metallic, using the rule that n − m must divide by 3. Then check each one.
2. Try (10,1), (11,2) and (12,3). They all have the same n − m. How does the gap change as the tube gets wider?
3. Find a semiconducting tube about 1.5 nm across. Is its gap bigger or smaller than silicon's 1.12 eV?
4. Look at where the red lines cross the blue cone for a metallic tube and for a semiconducting one. What is different?

Questions (check your answers on the page, under *Check yourself*)

- Why does a cutting line through K make the tube metallic?
- Why does the gap of a semiconducting tube scale as 1/d?
- Why is the density of states of a metallic tube flat and non-zero at E = 0?

2 The ballistic nanotube transistor



Link: https://faizverse.github.io/nanoelectronic-devices/L4/cntfet-ballistic.html
Read first: Document 2, Sections 3–4

What it shows. In a short nanotube, electrons cross from source to drain without scattering. The current is set by how many states at the top of the barrier are filled from each end. Change the gate, the drain, the tube and the insulator, and watch the quantum capacitance limit the charge.

Left. The energy of the first subband at the top of the barrier in the channel (black line), set by the gate. Electrons moving towards the drain come from the source and fill states up to the source Fermi level μ_S; those moving back come from the drain, filled up to $\mu_D = \mu_S - qV_D$. The current is the difference.

Right. The transfer curve. Below threshold the gate lowers the barrier and the current rises tenfold every 60 mV at 300 K. Above threshold the current rises roughly linearly, at most $4q^2/h = 155$ μA per volt for one subband.

The quantum capacitance. Adding electrons to the tube costs energy because its density of states is finite. With a thin high-κ gate, the gate moves the bands almost one-for-one, and the charge, and so the current, is set by the tube's own states.

> **The big idea.** In a short tube electrons cross without colliding. The gate sets how many of them can reach the top of the barrier, and because the tube has so few states, its own quantum capacitance limits the charge.

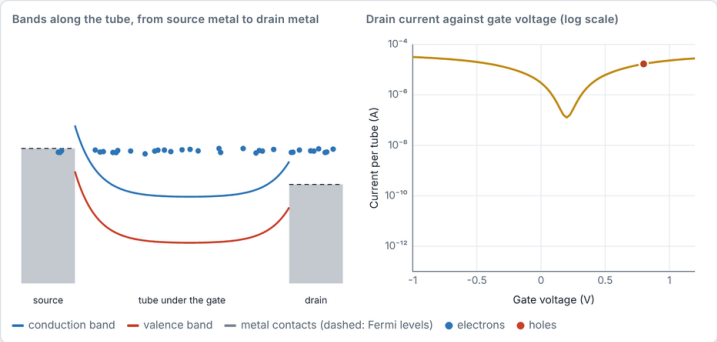
Try this

1. Sweep the gate at a drain voltage of 0.5 V. At low current, how much gate voltage does each tenfold rise in current need?
2. **Predict** what heating to 400 K does to that number. Check.
3. Change the insulator from SiO_2 to HfO_2, then thin it from 10 nm to 2 nm. Does the current keep rising as fast?
4. Set the drain voltage to 0.05 V, then raise it slowly. Where does the current stop growing?

Questions (check your answers on the page, under *Check yourself*)

- Why does the current saturate without pinch-off?
- Why does making the insulator thinner eventually stop helping?
- What is the most current one subband can carry?

3 Schottky-barrier nanotube transistor



Link: https://faizverse.github.io/nanoelectronic-devices/L4/sb-cntfet.html
Read first: Document 2, Sections 2 and 4.4

What it shows. With metal source and drain, the gate does not lower a barrier: it thins the Schottky barriers at the contacts so that electrons, or holes, tunnel through. See why such transistors conduct both ways, why wide tubes leak, and what doped contacts change.

Left. At each metal contact the bands are pinned: the conduction band starts a Schottky-barrier height φ_{Bn} above the metal's Fermi level. Inside the tube the gate moves the bands. The bands bend from the contact to the channel over a screening length λ, set by the gate insulator. A thin barrier can be tunnelled through.

Ambipolar. A high gate voltage pulls the conduction band down, thinning the barrier for **electrons** at the source. A low gate voltage pushes the bands up, thinning the barrier for **holes** at the drain. With midgap contacts both happen, and the transfer curve is V-shaped.

Doped contacts replace the metal–tube barrier by n^+ extensions: the device becomes a MOSFET-like, one-way transistor whose gate lowers a barrier in the channel.

> **The big idea.** With metal contacts the gate does not lower a barrier; it makes the barriers at the contacts thin enough to tunnel through. Electrons get through when the gate is high, holes when it is low.

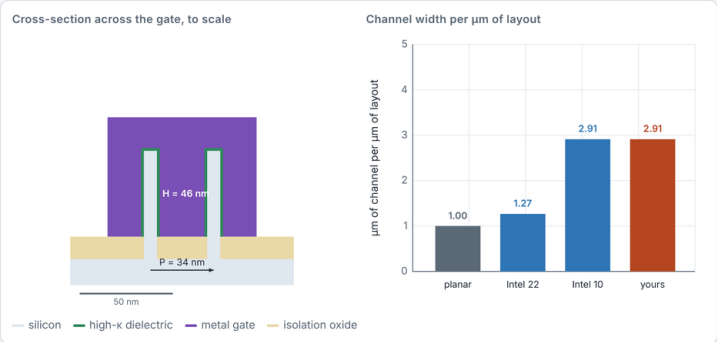
Try this

1. With midgap contacts, sweep the gate from −1 to +1.2 V. Which carriers flow at each end of the curve?
2. **Predict** what a wider tube (2.0 nm) does to the smallest current. Check, then try 1.0 nm.
3. Make the screening length λ longer, as a thicker gate insulator would. What happens to the steepness of the curve?
4. Choose doped contacts. Why does the curve no longer turn up at low gate voltage?

Questions (check your answers on the page, under *Check yourself*)

- What does the gate change in a Schottky-barrier transistor?
- Why do wide tubes leak more?
- Why does a thin, high-κ insulator help?

4 FinFET and nanosheet: width and control



Link: https://faizverse.github.io/nanoelectronic-devices/L4/finfet-nanosheet.html
Read first: Document 3, Sections 1–2

What it shows. Design a FinFET or a stacked-nanosheet transistor. Change fin height, width and pitch, or the number and size of the sheets, and see how much channel width fits in each micrometre of layout, and how well the gate controls a short channel.

FinFET. Each fin conducts along both sidewalls and its top, so its effective width is $2H_{fin} + W_{fin}$. Fins sit on a pitch P, so the width packed into each µm of layout is $(2H_{fin} + W_{fin})/P$, which exceeds 1 when fins are tall and close together. Width comes only in whole fins.

Nanosheet. Each sheet is surrounded by the gate, so it conducts on all four faces: 2W + 2T per sheet. The sheet width W is a free design choice, so there is no "whole fin" restriction, and stacking three or four sheets gives a lot of width from one footprint.

Control. A thin fin or sheet, gated from two or four sides, has a short natural length λ, so short gates still switch steeply. More detail shows the swing and DIBL from the same model as Lecture 1.

> **The big idea.** A fin conducts down both sides and over the top, so tall, closely packed fins give much more channel width than the flat chip area they sit on. Nanosheets stack the channels and let the width be chosen freely.

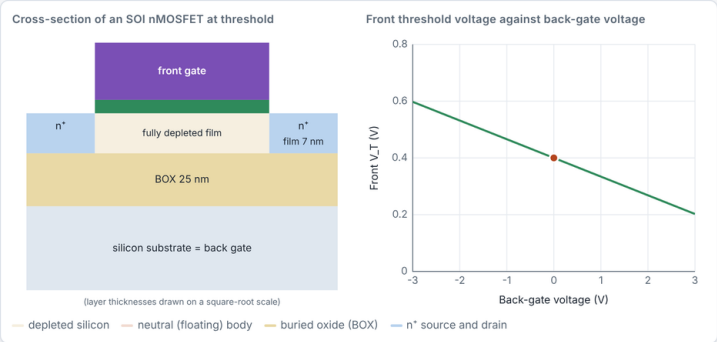
Try this

1. Load the Intel 22, 14 and 10 nm presets in turn. Which changed most: fin height, fin width or pitch?
2. **Predict** the fin height needed for 3 µm of channel per µm of layout at a 30 nm pitch. Check.
3. Switch to a nanosheet and change the sheet width. Why might a designer want this freedom?
4. In More detail, set a 12 nm fin with a 14 nm gate, then thin the fin to 6 nm. What happens to the swing?

Questions (check your answers on the page, under *Check yourself*)

- How can a FinFET have more width per µm than a planar transistor?
- Why does a thinner fin give better short-channel control?
- What problem of FinFETs does the nanosheet solve?

5 SOI: partially and fully depleted



Link: https://faizverse.github.io/nanoelectronic-devices/L4/soi-pd-fd.html
Read first: Document 3, Sections 6–7

What it shows. Thin the silicon film or change its doping and watch the depletion region reach the buried oxide: partially depleted becomes fully depleted. Then use the substrate as a second gate, and see the kink that a floating body causes.

Left. At threshold the gate depletes the film to a depth $x_{d,max} = \sqrt{(4\varepsilon_{Si}\varphi_F/qN_A)}$. If the film is thicker, a neutral body is left underneath, connected to nothing: **partially depleted** (PD). If it is thinner, the depletion reaches the BOX: **fully depleted** (FD).

Right. In an FD film the front and back interfaces are coupled through the film, so the substrate acts as a second gate. Its voltage shifts the front threshold along the Lim–Fossum line, until the back surface inverts or accumulates and screens it. In a PD film the neutral body screens the back gate completely.

More detail. In a PD device, holes from impact ionisation collect in the floating body, raise its potential and lower V_T: the current jumps, the "kink".

> **The big idea.** If the silicon film is thicker than the gate's depletion depth, a floating neutral layer is left: partially depleted. If it is thinner, the whole film is depleted, and the substrate below the buried oxide acts as a second gate.

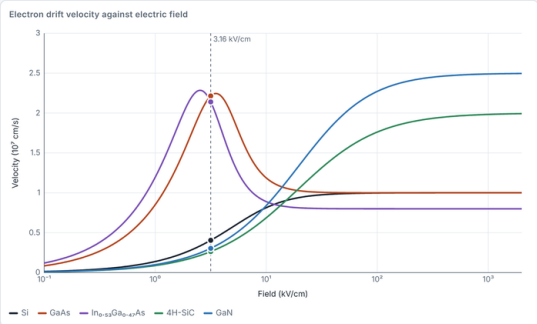
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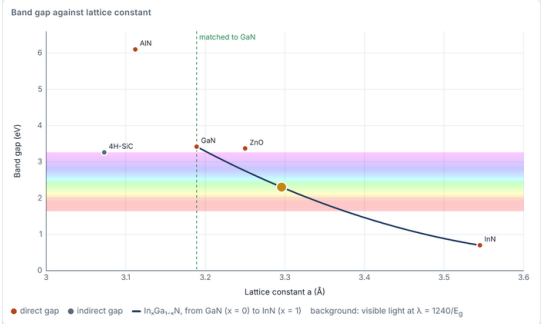
1. Start with the PD-SOI preset. Thin the film slowly. At what thickness does it become fully depleted?
2. Choose the 28 nm FD-SOI preset and sweep the back gate. By how many mV does the threshold move per volt?
3. **Predict** what a thick (145 nm) BOX does to that shift. Check.
4. In More detail, use a PD device, raise the drain voltage and find the kink. What does a body tie change?

Questions (check your answers on the page, under *Check yourself*)

- What is the condition for full depletion?
- Why does the back gate shift the threshold only in FD-SOI?
- Why does a thin BOX give stronger coupling?

6 Compound semiconductors: speed and colour





Link: https://faizverse.github.io/nanoelectronic-devices/L4/compound-semiconductors.html
Read first: Document 4, Sections 1–2

What it shows. Two views of why engineers use materials other than silicon. Compare how fast electrons move in Si, GaAs, InGaAs, SiC and GaN as the field rises; then move along an alloy and read off its band gap, lattice match and the colour of light it would emit.

Velocity–field. At low fields the velocity is mobility × field, so light-electron materials (GaAs, InGaAs) are far faster than silicon. In GaAs and InGaAs the velocity **peaks** at a few kV/cm and then falls: hot electrons move into heavy "satellite" valleys (the transferred-electron, or Gunn, effect). At high fields everything saturates, and the wide-gap materials SiC and GaN saturate highest.

Band-gap map. Alloys of two compounds have band gaps and lattice constants between those of their ends, so the gap can be tuned. A layer grows without defects only on a substrate with nearly the same lattice constant. A direct gap emits light at $\lambda \approx 1240/E_g$ nm.

> **The big idea.** Light electrons make GaAs and InGaAs fast at low fields, until hot electrons jump into heavy valleys. Mixing two compounds tunes the band gap, but layers grow cleanly only on a substrate with the same atom spacing.

Try this

1. Velocity tab: read each material's speed at 1 kV/cm and at 100 kV/cm. Does the order change?
2. Find the field at which GaAs is fastest. What happens at ten times that field, and why?
3. Map tab: move along InGaAs and find the mix that matches InP. What wavelength of light would it detect?
4. **Predict** where AlGaAs stops having a direct gap. Check, then find the InGaN mix that gives green light.

Questions (check your answers on the page, under *Check yourself*)

- Why does the GaAs velocity fall above about 4 kV/cm?
- Why can AlGaAs be grown on GaAs at any composition?
- Why do wide-gap materials matter for power electronics?